

QUESTION 1

One of the application areas of database systems is e-commerce such as online shopping.

What are the data or information stored in the database system of an online shopping website?
Give FOUR (4) of them.

Data	1. Order 2. Payment 3. Product 4. Customer
Information	1. Tupperware 2. Credit Card 3. Tupperware Bottle 4. Phone Number

QUESTION 2

Answer the questions below based on the following relational schema:

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SCIENTIST (Sc_ID, Sc_Name, Sc_Phone, Sc_Add)
EXPERIMENT (Exp_NO, Ex_Name, Proj_Code)
PROJECT (Proj_Code, Proj_Name, Proj_StartDate)
SCIENTIST_EXPERIMENT (Sc_ID, Exp_NO, Date, Time)
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- a) For each table, identify the primary key and foreign key. If a table does not have a primary key or foreign key, write – in the space provided. (4M)

TABLE	PRIMARY KEY	FOREIGN KEY
SCIENTIST	Sc_ID	-
EXPERIMENT	Exp_NO	Proj_Code
PROJECT	Proj_Code	-
SCIENTIST_EXPERIMENT	Time	Sc_ID, Exp_No

- b) Does the table exhibit entity integrity? Answer 'Yes' or 'No' and then explain your answer. (4M)

TABLE	ENTITY INTEGRITY	EXPLANATION
SCIENTIST	Yes	It has Primary Key, which is Sc_ID
EXPERIMENT	Yes	It has Primary Key, which is Exp_NO
PROJECT	Yes	It has Primary Key, which is Proj_Code

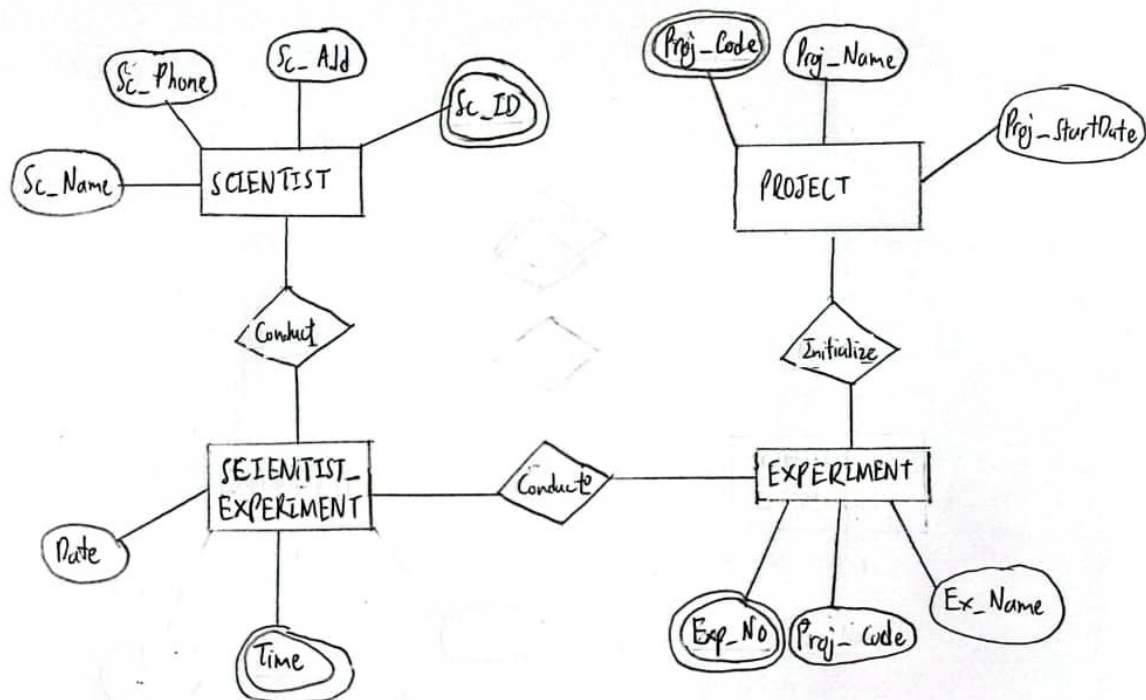
SCIENTIST_EXPERIMENT	Yes	It has Primary Key, which is Time
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- c) Does the table exhibit referential integrity? Answer 'Yes' or 'No' and then explain your answer. (4M)

TABLE	REFERENTIAL INTEGRITY	EXPLANATION
SCIENTIST	No	It has no Foreign Key
EXPERIMENT	Yes	It has Foreign Key, which is Proj_Code
PROJECT	No	It has no Foreign Key
SCIENTIST_EXPERIMENT	Yes	It has Foreign Key, which are Sc-ID and Exp_No

- d) Draw ERD using Chen notation based on the given relational schema. (4M)

d)



QUESTION 3

Read the following and complete the task listed.

Consider the following about Luxury Records:

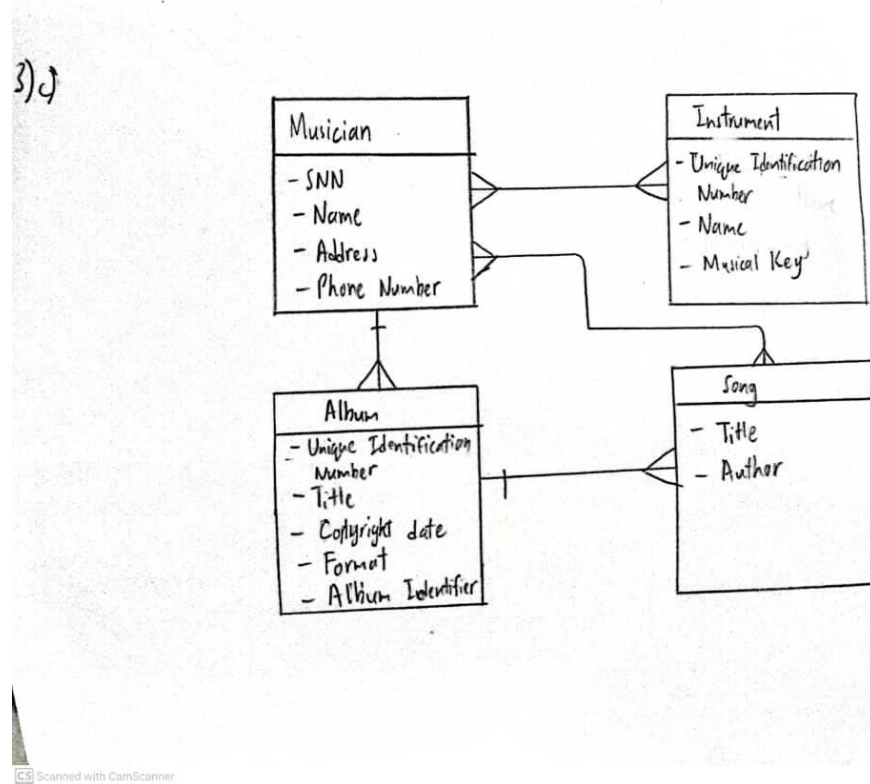
Here is the information that you gather:

- Luxury has decided to store information about musicians who perform on its albums (as well as other company data) in a database.
- Each musician that records at Luxury has an SSN, a name, an address, and a phone number.
- Poorly paid musicians often share the same address, and no address has more than one phone.
- Each instrument used in songs recorded at Luxury has a unique identification number, a name (e.g guitar, synthesizer, and a flute) and a musical key (e.g., C, B-flat, E-flat).
- Each album recorded on the Luxury label has a unique identification number, a title, a copyright date, a format (e.g., CD or MC) and an album identifier.
- Each song recorded at Luxury has a title and an author.
- Each musician may play several instruments, and a given instrument may be played by several musicians.
- Each song album has a number of songs on it, but no song may appear on more than one album.
- Each song is performed by one or more musicians, and a musician may perform a number of songs.
- Each album has exactly one musician who acts as its producer.
- A musician may produce several albums, of course.

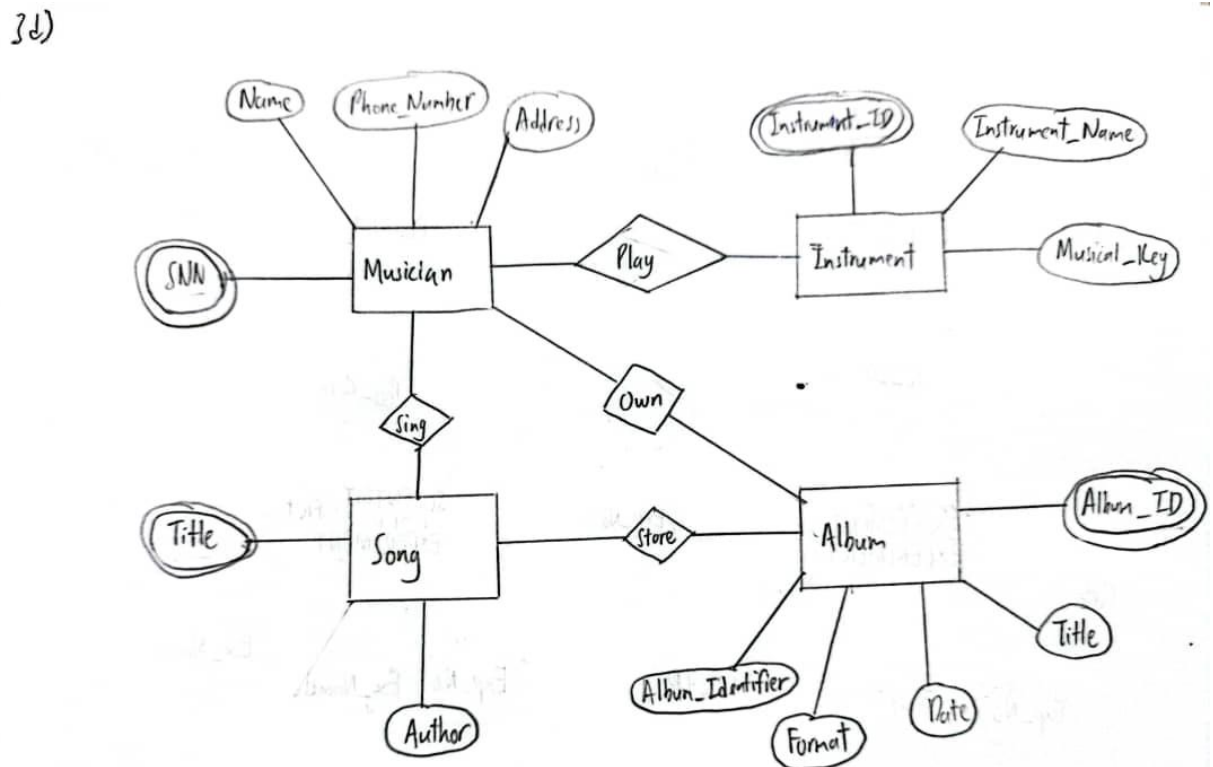
Task:

1. List each Entity that is identified in Luxury Records System.
 - a) **Musician**
 - b) **Instrument**
 - c) **Album**
 - d) **Song**
2. For each entity list the attributes.
 - a) **Musician (SSN, Name, Address, Phone Number)**
 - b) **Instrument (Instrument_ID, Instrument_Name, Musical Key)**
 - c) **Album (Album_ID, Title, Date, Format, Album_Identifier)**
 - d) **Song (Song_ID, Title, Name)**

3. Draw a logical model to represent each entity and the relationships between them.



4. Draw an Entity Relationship Diagram (ERD) that captures the entire system.



QUESTION 4

Give an example of each of the three types of relationships.

- i) One-to-many relationship
 - **A lecturer can have multiple students, but student can only choose one lecturer in one course**
- ii) Many-to-many relationship
 - **Gamer can own multiple games, while one game can be owned by multiple gamers**
- iii) One-to-one relationship
 - **One student can only have one dome**
- iv) Many-to-one relationship
 - **Many staffs work under one company.**

QUESTION 5

CLIENT (Client_Number, Clent_Name, Street, City, State, Postal_Code, Amount_Paid, Current_Due, Recruiter_Number)

RECRUITER (Recruiter_Number, Last_Name, First_Name, Street, City, State, Postal_Code, Rate, Commission)

- a) What is the relationship between Recruiter and Client?
 - **Many to One**
- b) What is the primary key for the Client table and Recruiter table?
 - **Client_Number**
 - **Recruiter_Number**
- c) Define foreign key? Identify foreign key in the Client table.
 - **A foreign key is an attribute in one table that refers to a primary key in another table. Its purpose is to establish relationship between one table to another.**
 - **Recruiter_Number**

QUESTION 6

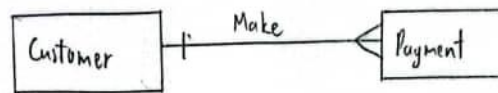
Draw the Chen & Crow's Foot notation based on the following statements:

- i) "A customer can make many payments, but each payment is made by only one customer."
- ii) "A medical doctor may treat many patients. Some patients may be treated by more than one medical doctor. A patient may have or may not have insurance plan."

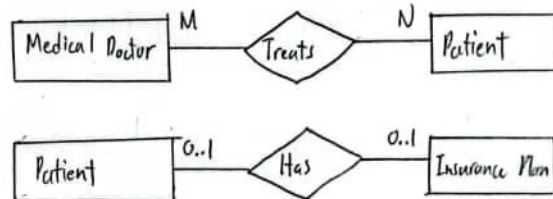
i). Chen Notation



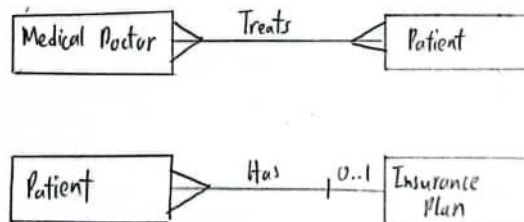
• Crow's Foot Notation



ii). Chen Notation



• Crow's Foot Notation



QUESTION 7

Consider the following relations for a database that keeps track of student enrollment in courses and the books adopted for each course:

STUDENT(SSN, Name, Major, Bdate)

COURSE(Course#, Cname, Dept)

ENROLL(SSN, Course#, Quarter, Grade)

BOOK_ADOPTION(Course#, Quarter, Book_ISBN)

TEXT(Book_ISBN, Book_Title, Publisher, Author)

Draw a relational schema diagram specifying the foreign keys for this schema.

7)

